

AWEF Net: An Attention Weighted Ensemble Fusion Framework for Multi Class Pulmonary Disease Classification from Chest Radiographs

Abstract—Heterogeneous convolutional ensembles are widely used to improve pulmonary disease classification from chest radiographs (CXRs). However, common late fusion schemes, such as direct feature concatenation or equal weight prediction averaging, use combination rules that do not adapt to the input image. This can create an inter model feature conflict in which one backbone dominates the fused decision and limits the contribution of complementary models. We present AWEF Net, an Attention Weighted ensemble fusion framework that combines ResNet50, InceptionV3, and DenseNet201 through a lightweight two layer multilayer perceptron attention gate. The gate produces input adaptive scalar weights for the three backbones and contributes fewer than 0.6% of the total model parameters. On the COVID-19 Radiography Database, containing 21,165 images from four classes, five fold stratified cross validation yields a macro-F1 score of 96.3% and a COVID-19 recall of 97.1%. These results exceed the strongest fixed fusion baseline by 1.4 and 1.7 percentage points, respectively. Ablation experiments identify adaptive gating as the largest contributor to the improvement, while evaluation on an external CXR corpus indicates better cross dataset generalisation than fixed fusion. Analysis of the learned weights further shows class dependent backbone reliance. Source code will be made publicly available upon publication.

Index Terms—Chest radiography, ensemble learning, attention mechanism, feature fusion, COVID-19, medical image classification, deep learning.

I. INTRODUCTION

Chest radiography remains one of the most accessible imaging modalities for examining pulmonary disease. In recent years, several studies published between 2024 and 2026 have continued to report strong performance in CXR image classification using lightweight CNNs, transformer-CNN hybrid models, and attention based methods [1]–[3]. As different architectures represent local, multiscale, and global features in different ways, modern ensemble and fusion approaches increasingly use multiple backbone models instead of relying on a single model [4]–[6]. In practice, these ensembles are often constructed through late fusion, either by concatenating global average pooled feature vectors or by averaging model predictions.

These fixed fusion rules may not fully exploit backbone complementarity. With feature concatenation, a shared classifier learns a fixed block of weights for each backbone; the relative contribution of those blocks does not change from one radiograph to another. Differences in feature scale and representation can therefore cause one backbone to dominate the logits or optimisation dynamics, reducing the influence of other models. We refer to this imbalance as *inter model feature*

conflict. Recent ensemble studies on CXRs have reached similar conclusions regarding the robustness benefits of uncertainty aware or image adaptive weighting over static model combinations [4], [5]. The problem is particularly relevant when ensemble diversity is expected to improve recall for clinically important classes such as COVID-19.

This work investigates whether a small trainable gate can reduce this conflict by assigning input dependent importance to each backbone. We propose **AWEF Net** (Attention Weighted Ensemble Fusion Network), which first projects the features from ResNet50, InceptionV3, and DenseNet201 into a common space. A two layer multilayer perceptron (MLP) then generates per backbone weights ($\alpha_1, \alpha_2, \alpha_3$) for each input radiograph. The gate is differentiable and lightweight, allowing the backbone contributions and final classifier to be optimised jointly.

The main contributions of this study are listed below:

- Feature conflict among different CXR models is discussed as a limitation of fixed late fusion in heterogeneous ensembles.
- An Attention Weighted ensemble framework, called AWEF Net, is proposed and validated with reproducible five fold stratified cross validation on the COVID-19 Radiography Database.
- The impact of adaptive attention weighting, backbone diversity, differential learning rates, and label smoothing is isolated through ablation studies.
- Zero shot generalisation and the dependence of different classes on particular backbones are explored through the learned attention weights.

II. RELATED WORK

A. Transfer Learning and Lightweight CXR Models

ImageNet pretrained CNNs have been widely adopted in CXR analysis because they provide reliable transfer learning baselines at a practical computational cost. In particular, ResNet, InceptionV3, and DenseNet offer residual, multiscale, and densely connected feature extraction, respectively [7]–[9]. Recent approaches have focused on maintaining high performance while reducing inference cost. Yen and Tsao proposed a lightweight multiscale CNN for COVID-19 and pneumonia detection [1], while Maurya *et al.* designed SqueezeViT to decrease the number of tokens and memory consumption in transformers used for CXR classification [10]. Oltu *et al.* combined DenseNet201 and Vision Transformer representations to

recognise four classes of pulmonary disease [2]. These works show that architecture selection affects both representation ability and model deployability, motivating the use of diverse backbone architectures in AWEF Net.

B. Large Scale and Foundation Models

Another stream of research focuses on scaling and diversifying CXR pretraining. Cid *et al.* introduced open source networks pretrained on multicentre medical data and evaluated them externally across various tasks [11]. The self supervised EVA-X framework was applied to more than half a million radiographs and produced representations applicable to classification, localisation, and segmentation [12]. Other studies have incorporated global transformer and local CNN features through hybrid architectures, including the ViT-DenseNet architecture of Oltu *et al.* [2] and the multiscale ViT-DenseNet cross attention approach of Shi *et al.* [6]. Unlike large or tightly coupled hybrid architectures, AWEF Net uses standard pretrained backbones with a small and interpretable fusion gate.

C. Ensemble Learning and Adaptive Fusion

Classic approaches to CXR ensembles typically use majority voting, probability averaging, or feature concatenation across independently trained models. Such methods can improve average accuracy but apply the same fusion structure to all images. Abad *et al.* addressed variability across different data sources through uncertainty weighting based on the entropy of heterogeneous CNN predictions and demonstrated better external performance than individual models [4]. Thakare *et al.* proposed image specific, entropy regularised fusion between CNN and transformer streams [5]. Shi *et al.* used bidirectional cross attention to align convolutional feature maps and transformer tokens across several scales [6]. These approaches support adaptive fusion, but they either weight final predictions or fuse two closely coupled streams. In contrast, AWEF Net estimates three scalar backbone weights from projected penultimate features, allowing direct comparison of heterogeneous model outputs with low parameter overhead.

D. Attention and Pathology Relevant Representation

Common uses of attention mechanisms in CXR analysis include suppressing irrelevant anatomical structures, identifying lesion locations, and improving interactions between feature streams. Mahawar and Paul created disease specific maps using attention and increased cross dataset performance through prediction self consistency [13]. Oltu *et al.* incorporated an attention mechanism into transformer based multiclass lung disease classification [2]. Imran *et al.* investigated lightweight spatial attention modules for multilabel thoracic pathology detection across several CNN backbones [3]. Wu *et al.* extended attention by integrating it with lung segmentation and used a gradient based attention objective to focus learning on clinically relevant regions [14]. Unlike these approaches, AWEF Net does not claim to perform lesion localisation. Instead, its attention mechanism operates at the model level:

the gate determines how much weight should be assigned to each complete backbone.

E. Domain Shift and External Validation

High internal performance scores do not necessarily imply that a model will generalise to images from other institutions or populations. Hoebel *et al.* identified systematic, pathology-dependent prediction differences in models trained on major CXR databases [15]. Current strategies for addressing domain shift include Wasserstein-based feature alignment [16], supervised adversarial adaptation across patient populations [17], and attention-guided disease decomposition [13]. Large Scale multicentre research has also highlighted the importance of testing on independent sources: Cid *et al.* evaluated their open source model on external radiographs [11], while Liu *et al.* trained and validated a multi diagnosis system using images collected across several countries [18]. These studies motivate the external Zero shot test used in this paper and reinforce that neither internal cross validation nor attention weighting alone should be interpreted as evidence of clinical readiness.

F. Research Gap and Positioning of AWEF Net

The literature reveals three distinct levels at which adaptive weighting can be applied. Prediction level ensembles combine class probabilities after each model makes an independent decision [4]. Representation level hybrids exchange or align intermediate CNN and transformer features through attention [5], [6]. spatial attention methods alter where a single network focuses within a radiograph [3], [13], [14]. These techniques are related but do not completely overlap: output weighting cannot directly resolve mismatched feature scales, tightly coupled cross attention increases architectural complexity, and spatial attention does not decide which complete backbone should dominate a case.

AWEF Net addresses the more specific problem of model level allocation within a heterogeneous CNN ensemble. It reduces dimensional incompatibility before fusion through common projections and constrains the backbone coefficients to the probability simplex through a softmax gate. The architecture therefore combines the independent inductive biases of ResNet50, InceptionV3, and DenseNet201 without requiring a transformer branch, lesion annotations, or a large foundation-model pretraining stage. The present evaluation also separates the contribution of the gate from architectural diversity and optimisation choices. This positioning is important because improvements from a complex ensemble could otherwise be attributed to any of these components. The external Zero shot test and class wise weight analysis are consequently used as complementary evaluations, not as substitutes for prospective clinical validation.

III. METHODOLOGY

A. Problem Formulation and Feature Conflict

Let x denote a chest radiograph and $\mathcal{B} = \{B_1, B_2, B_3\}$ denote ImageNet pretrained ResNet50, InceptionV3, and DenseNet201 backbones [7]–[9]. Each backbone produces a

global average pooled feature vector $f_k = \text{GAP}(B_k(x)) \in \mathbb{R}^{d_k}$, where $d_1 = 2048$, $d_2 = 2048$, and $d_3 = 1920$. Naive feature level fusion concatenates the vectors into $z = [f_1; f_2; f_3]$ and applies a linear classifier $W = [W^{(1)} \ W^{(2)} \ W^{(3)}]$, producing

$$s(x) = \sum_{k=1}^3 W^{(k)} f_k. \quad (1)$$

Although the classifier weights are trainable, their relationship to the three backbones is fixed after training and does not adapt to x . If $\|f_j\| \gg \|f_k\|$ for some pair (j, k) , or if one block produces consistently larger logit contributions, backbone j can dominate the decision and associated gradients. Complementary evidence from backbone k may then be underused. AWEF Net is designed to make this relative contribution input dependent. The GAP module summarizes the spatial activation tensor of each backbone into a small channel-wise tensor, such that the fusion step can compare the resulting representation of each architecture without retaining the complete set of feature maps. This architecture further allows placing the fusion decision prior to the classifier, hence making both the backbone choice and class discrimination possible using a single end-to-end objective.

B. AWEF Net Architecture

AWEF Net replaces the fixed combination in (1) with the adaptive fusion mechanism shown in Fig. 1. First, each backbone feature is projected into a common $d = 1024$ dimensional space:

$$\hat{f}_k = W_k^p f_k + b_k^p, \quad \hat{f}_k \in \mathbb{R}^{1024}. \quad (2)$$

The projected features are concatenated as $g = [\hat{f}_1; \hat{f}_2; \hat{f}_3] \in \mathbb{R}^{3072}$ and passed through a two layer MLP followed by softmax normalisation. Using equal-width projections in all branches ensures that the backbone outputs do not suffer from inherent dimension mismatch and makes their projections consistent and comparable to one another. As the gate operates on the concatenated vector as a whole, each attention value becomes dependent on the information present in all three branches.

$$a = W_2 \sigma(W_1 g + b_1) + b_2, \quad a \in \mathbb{R}^3, \quad (3)$$

$$\alpha_k = \frac{\exp(a_k)}{\sum_{j=1}^3 \exp(a_j)}, \quad \sum_{k=1}^3 \alpha_k = 1, \quad (4)$$

where σ is the ReLU activation and $W_1 \in \mathbb{R}^{128 \times 3072}$. Since the softmax normalization forces the coefficients to be between zero and one and to sum up to one, the bounded fusion is ensured, and the weights can be understood as a measure of relative contribution of each backbone. The fused representation is the convex combination

$$f_{\text{fused}} = \sum_{k=1}^3 \alpha_k \hat{f}_k \in \mathbb{R}^{1024}. \quad (5)$$

A compact classification head then produces the four class prediction:

$$h = \text{Drop}(\text{BN}(\sigma(W_h f_{\text{fused}}))), \quad (6)$$

$$\hat{y} = \text{softmax}(W_c h), \quad (7)$$

where $W_h \in \mathbb{R}^{512 \times 1024}$, the dropout rate is 0.4, and $W_c \in \mathbb{R}^{4 \times 512}$. Because α depends on x , the ensemble can assign different relative importance to the three backbones for different radiographs. The weighted sum keeps the resulting vector to have the size of 1024, as required by the classification head, thus eliminating the increase in dimension that would have happened in the case of simple concatenation. During backpropagation, the classifier loss simultaneously updates the projections, the attention gate, and the classifier itself, encouraging the selection of the branches that contain valuable information about the current sample.

C. Training Objective

The model is trained with label smoothed cross entropy. For a one-hot target y over $C = 4$ classes and smoothing coefficient ϵ ,

$$\tilde{y}_c = (1 - \epsilon)y_c + \frac{\epsilon}{C}, \quad \mathcal{L} = - \sum_{c=1}^C \tilde{y}_c \log \hat{y}_c. \quad (8)$$

The pretrained backbones are fine-tuned with a lower learning rate than the newly initialised projection, gate, and classification modules. This differential learning rate strategy limits abrupt changes to transferred representations while allowing the fusion layers to adapt more rapidly. Label smoothing helps to relax the requirement to assign the full probability to a single target class, which regularizes training when the appearance of different diseases overlaps in radiographs. The same loss function guides the class prediction as well as the backbone weight allocation.

IV. EXPERIMENTAL SETUP

A. Datasets

The primary corpus is the COVID-19 Radiography Database [1], [2], [19], which contains 21,165 posterior-anterior chest radiographs from four classes (Table I). The class distribution is imbalanced, with a Normal to Viral Pneumonia ratio of approximately 7.6:1. Stratified folds preserve the class proportions, while augmentation and label smoothing regularise model training.

For external domain evaluation, we use the public Chest X-Ray (Pneumonia, COVID-19, Tuberculosis) corpus [20]. The complete corpus contains 7,135 images from four categories. Tuberculosis images are excluded, leaving 6,406 images from COVID-19, Normal, and Pneumonia for evaluation. The source model outputs corresponding to COVID-19, Normal, and Viral Pneumonia are used for this three class comparison, with Viral Pneumonia treated as the source domain counterpart of the external Pneumonia category. No external images are used for fine tuning or model selection.

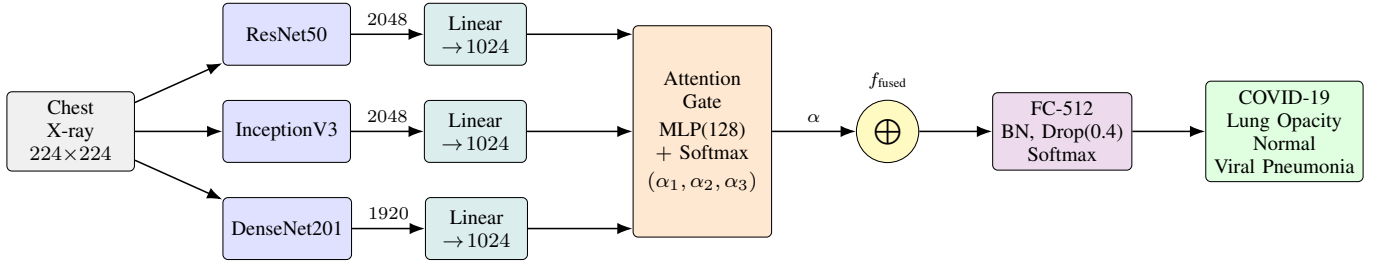


Fig. 1. AWEF Net projects three backbone features, learns input adaptive weights with an MLP gate, and sums them for four class prediction.

TABLE I
COMPOSITION OF THE PRIMARY DATASET

Class	Images	Share (%)
COVID-19	3,616	17.1
Lung Opacity	6,012	28.4
Normal	10,192	48.2
Viral Pneumonia	1,345	6.4
Total	21,165	100.0

TABLE II
FIVE FOLD CROSS VALIDATION ON THE PRIMARY DATASET (%). MEAN $\pm$ STANDARD DEVIATION IS SHOWN WHERE AVAILABLE.

Model	Acc.	Macro-F1	AUC	COVID Rec.
ResNet50	94.8 $\pm$ 0.5	93.6 $\pm$ 0.5	99.1	93.9 $\pm$ 0.6
InceptionV3	93.9 $\pm$ 0.6	92.5 $\pm$ 0.6	98.8	92.8 $\pm$ 0.7
DenseNet201	95.3 $\pm$ 0.4	94.2 $\pm$ 0.5	99.3	94.7 $\pm$ 0.5
Feature Concatenation	95.5 $\pm$ 0.4	94.5 $\pm$ 0.4	99.3	95.1 $\pm$ 0.5
Probability Average	95.8 $\pm$ 0.4	94.9 $\pm$ 0.4	99.4	95.4 $\pm$ 0.4
AWEF Net	97.1$\pm$0.3	96.3$\pm$0.4	99.6	97.1$\pm$0.4

B. Implementation Details

Images are converted to three channels and normalised using ImageNet statistics. ResNet50 and DenseNet201 receive 224×224 images, while the InceptionV3 branch is internally resized to 299×299 . Training time augmentation consists of random horizontal flipping, brightness and contrast jitter of up to $\pm 10\%$, and random resized cropping.

Optimisation relies on AdamW with learning rates of 1×10^{-4} for the pretrained backbones and 1×10^{-3} for the projection, gate, and classification heads. The weight decay is 1×10^{-4} , the batch size is 32, and the label smoothing coefficient is $\epsilon = 0.1$. Training runs for at most 50 epochs with early stopping using a patience of 10 epochs and a ReduceLROnPlateau schedule. We report five fold stratified cross validation with seed 42. Identical fold indices are used for all models to support paired comparison. The attention gate contains approximately 0.39 million parameters, corresponding to about 0.55% of the complete model.

C. Evaluation Metrics

Macro averaged F1 is the primary metric because it gives equal importance to each class under imbalance. We additionally report overall accuracy, macro one vs rest area under the receiver operating characteristic curve (AUC), and class-specific precision, recall, and F1. COVID-19 recall is highlighted because false negative predictions are especially undesirable in a screening setting. Reported cross validation results aggregate the five held out folds.

V. RESULTS AND DISCUSSION

A. Main Comparison

Table II compares AWEF Net with the individual backbones and two fixed fusion baselines. AWEF Net achieves the highest

TABLE III
PER CLASS METRICS FOR AWEF NET (%)

Class	Precision	Recall	F1
COVID-19	97.6	97.1	97.3
Lung Opacity	96.7	95.4	96.1
Normal	97.7	98.4	98.0
Viral Pneumonia	92.6	94.6	93.6
Macro	96.2	96.4	96.3

mean macro-F1 (96.3%) and accuracy (97.1%). Feature concatenation and probability averaging improve only modestly over the strongest individual backbone, DenseNet201, whose macro-F1 is 94.2%. In contrast, adaptive fusion improves macro-F1 by a further 1.4 percentage points over probability averaging. COVID-19 recall reaches 97.1%, which is 1.7 points higher than the strongest fixed fusion baseline. These results indicate that an input dependent combination can exploit backbone complementarity more effectively than the evaluated fixed rules.

B. Per Class Analysis

Table III reports Per Class performance, and Fig. 2 shows the confusion matrix aggregated across the five held out folds. The strongest F1 is obtained for the majority Normal class (98.0%), while the minority Viral Pneumonia class reaches 93.6% despite accounting for only 6.4% of the dataset. The largest off diagonal count occurs between Lung Opacity and Normal, indicating that this class pair remains the main source of residual error.

		Predicted class			
		COVID	Lung Op.	Normal	Viral Pn.
True class	COVID	3511	48	22	35
	Lung Op.	40	5735	195	42
	Normal	18	120	10029	25
	Viral Pn.	30	25	18	1272

Fig. 2. Confusion matrix aggregated over the five held out folds. Counts sum to the complete 21,165-image primary dataset.

TABLE IV
ABLATION STUDY USING MACRO-F1 (%). Δ IS RELATIVE TO THE FULL MODEL.

Configuration	Macro-F1	Δ
AWEF Net (full)	96.3	—
Without adaptive gate (equal feature weights)	94.0	−2.3
Without backbone diversity (3×ResNet50)	94.6	−1.7
Without differential learning rates	95.5	−0.8
Without label smoothing	95.8	−0.5

C. Ablation Study

To determine which components account for the performance gain, each design choice is removed individually and the resulting model is trained under the same protocol (Table IV). Replacing the attention gate with equal weight averaging of the projected features produces the largest decrease, reducing macro-F1 by 2.3 points. This ablation differs from the probability averaging baseline in Table II: it retains the shared projection and classification head but fixes $\alpha_k = 1/3$ in Eq. (5). Replacing the heterogeneous ensemble with three ResNet50 branches produces the second largest decrease (1.7 points), supporting the value of architectural diversity. Differential learning rates and label smoothing provide smaller but consistent improvements.

D. external domain Generalisation

Table V reports Zero shot evaluation on the external three class corpus. AWEF Net achieves the highest accuracy (89.6%), macro-F1 (88.2%), and COVID-19 recall (90.4%). Relative to probability averaging, the gains are 4.8 points in macro-F1 and 5.5 points in COVID-19 recall. The result suggests that adaptive fusion is less sensitive than the evaluated fixed fusion rules to the distribution differences between the primary and external datasets.

E. Attention Weight Analysis

Figure 3 shows the mean gate weight assigned to each backbone for each class. The weights are not uniform. DenseNet201 receives the largest mean weight for COVID-19 and Viral Pneumonia, while ResNet50 receives the largest

TABLE V
ZERO SHOT EXTERNAL DOMAIN EVALUATION (THREE CLASSES, %)

Model	Acc.	Macro-F1	COVID Rec.
DenseNet201	83.7	81.8	82.6
Probability Average	85.1	83.4	84.9
AWEF Net	89.6	88.2	90.4

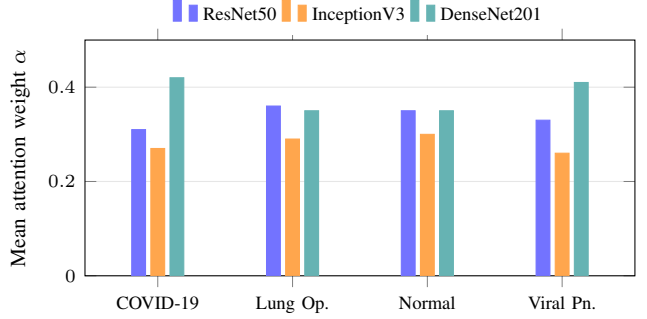


Fig. 3. class wise mean backbone weights learned by the gate. Per image weights sum to one, and bars summarise held out fold averages.

mean weight for Lung Opacity. ResNet50 and DenseNet201 receive equal mean weight for Normal in the reported results. InceptionV3 receives a lower average weight across all four classes, but remains part of the learned mixture and may contribute image specific evidence that is not visible from class means alone. This pattern demonstrates that the gate learns class dependent model reliance that fixed averaging cannot represent. The weights describe the model’s internal allocation of evidence and should not be interpreted as causal clinical explanations.

F. Discussion

The experiments consistently favour input adaptive fusion. The fixed fusion baselines provide modest improvements over the strongest individual backbone, whereas the attention gate produces a larger gain with limited parameter overhead. The ablation study indicates that adaptive gating is the most influential tested component, followed by heterogeneous backbone selection. The external domain results further suggest that the learned combination transfers better than the evaluated fixed rules. Such an issue aligns with current observations showing that CXR performance may differ systematically across datasets and patient groups, highlighting the need for domain adaptation and cross source validation [13], [15]–[17].

Several limitations should be considered. First, training uses a single primary dataset, and the external corpus is a composite public dataset rather than a prospective institutional cohort. Second, mapping the source Viral Pneumonia category to the external Pneumonia category introduces a label space mismatch. Third, the study does not include radiologist comparison, prospective validation, or statistical hypothesis testing across models. Finally, attention weights provide a useful description of model reliance but do not establish clinical causality. Future work should therefore evaluate multi source

training, patient level external cohorts, transformer based backbones, calibrated uncertainty, and prospective comparison with clinical readers.

VI. CONCLUSION

This paper introduced AWEF Net, an Attention Weighted fusion framework for heterogeneous CXR backbones. The method projects ResNet50, InceptionV3, and DenseNet201 features into a common space and uses a lightweight MLP gate to assign input adaptive backbone weights. In five fold stratified cross validation on the COVID-19 Radiography Database, AWEF Net achieves a macro-F1 score of 96.3% and a COVID-19 recall of 97.1%. Ablation results identify adaptive gating as the largest contributor to performance, while external domain evaluation shows higher macro-F1 and COVID-19 recall than fixed fusion. The learned weights also reveal class dependent backbone reliance. These findings support attention gated fusion as a practical alternative to fixed concatenation or averaging, while further multi source and prospective validation remains necessary before clinical use.

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